

Stroke Prediction Analysis

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Introduction

Health is one of the most important aspects of people's lives, and understanding the conditions that put individuals at risk can lead to earlier intervention and better outcomes. With strokes being one of the leading causes of disability and death worldwide, I wanted to explore what factors contribute most to stroke risk and how different patient characteristics compare. Using a medical dataset containing demographic information, lifestyle habits, and health indicators, I set out to analyze patterns among patients and identify warning signs. My goal was to clean and organize this data, uncover meaningful insights through exploratory analysis, and build predictive models that can help estimate a person's likelihood of experiencing a stroke. Through this work, I aim to make stroke risk factors more understandable and support data-driven healthcare decision-making.

Data Exploration

How The Data Was Collected

The dataset used in this project comes from Kaggle's publicly available Stroke Prediction Dataset, which was originally compiled from medical records. The dataset includes patient demographics, lifestyle information, and key health indicators often associated with stroke risk. Each record corresponds to a single patient and specifies whether the individual has previously experienced a stroke. Since the dataset is already structured and formatted for analysis, it provided a reliable foundation for both exploratory data analysis and predictive modeling.

Issues Encountered

A major challenge in this dataset is the class imbalance. Only about 5% of patients had a recorded stroke. This caused early machine learning models to predict that almost every patient was low-risk, resulting in poor detection of actual stroke cases. To address this, I applied SMOTE (Synthetic Minority Oversampling Technique) to oversample the minority class, which improved the model's ability to identify high-risk patients more accurately.

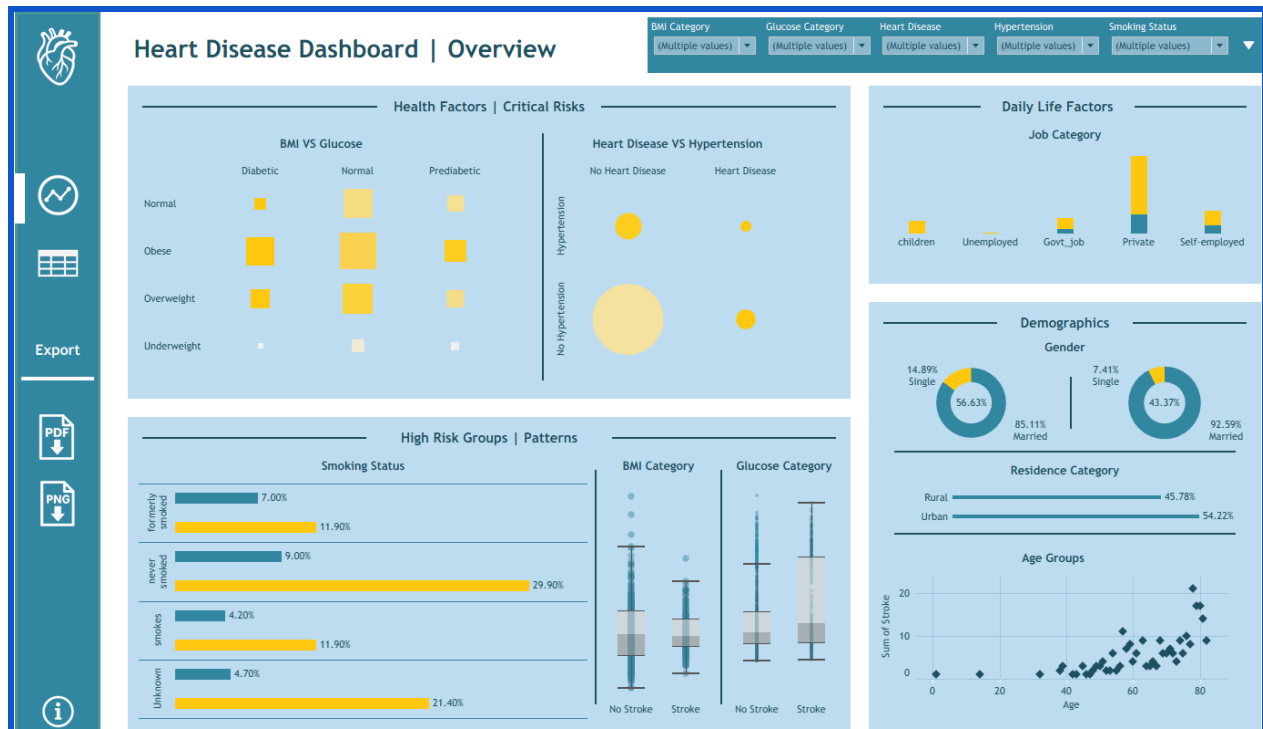
Features Identified For Analysis

This project focuses on a set of demographic, health, and lifestyle-related attributes that research has shown to be connected to stroke risk:

- Age
- Gender
- BMI
- Average glucose level
- Smoking status
- Hypertension
- Heart disease
- Marital status
- Work type
- Residence type

These features provide a broad understanding of both underlying medical conditions and behavioral contributors. By analyzing their relationship with stroke outcomes, the project aims to identify which factors have the strongest impact and how they interact to influence health risks.

Visualisations Generated Using Power Tableau



Heart Disease Dashboard | Overview

ID: 99 | Demographics: 31 | Single | Private | Urban | No Heart Disease | No Hypertension | Obese | Prediabetic | Unknown | No Stroke

ID: 129 | Demographics: 24 | Single | Private | Urban | No Heart Disease | No Hypertension | Overweight | Normal | never smoked | No Stroke

ID: 210 | Demographics: 81 | Married | Self-employed | Rural | No Heart Disease | No Hypertension | Obese | Normal | never smoked | Stroke

ID: 224 | Demographics: 23 | Single | Private | Urban | No Heart Disease | No Hypertension | Obese | Prediabetic | never smoked | No Stroke

ID: 491 | Demographics: 74 | Married | Self-employed | Urban | No Heart Disease | No Hypertension | Overweight | Normal | never smoked | Stroke

ID: 530 | Demographics: 12 | Single | children | Rural | No Heart Disease | No Hypertension | Normal | Normal | Unknown | No Stroke

ID: 545 | Demographics: 42 | Married | Private | Rural | No Heart Disease | No Hypertension | Obese | Diabetic | never smoked | No Stroke

ID: 621 | Demographics: 69 | Married | Private | Rural | No Heart Disease | No Hypertension | Overweight | Prediabetic | smokes | No Stroke

ID: 712 | Demographics: 82 | Single | Private | Rural | Heart Disease | Hypertension | Overweight | Normal | formerly smoked | Stroke

ID: 721 | Demographics: 52 | Married | Self-employed | Urban | No Heart Disease | Hypertension | Normal | Prediabetic | formerly smoked | No Stroke

ID: 820 | Demographics: 59 | Married | Private | Rural | No Heart Disease | Hypertension | Normal | Normal | never smoked | No Stroke

ID: 967 | Demographics: 61 | Married | Private | Urban | Heart Disease | No Hypertension | Obese | Normal | never smoked | No Stroke

ID: 1210 | Demographics: 68 | Married | Private | Rural | No Heart Disease | No Hypertension | Obese | Diabetic | Unknown | Stroke

ID: 1246 | Demographics: 43 | Married | Self-employed | Rural | No Heart Disease | No Hypertension | Obese | Prediabetic | formerly smoked | No Stroke

Methods

Loading the Dataset

The dataset was imported from Kaggle into Python using Pandas. Once loaded, the id column was removed since it does not contribute to prediction or analysis. The remaining medical and demographic features were used as inputs for exploratory analysis and machine learning.

Understanding the Dataset

The dataset contains patient-level health information including age, gender, BMI, average glucose level, smoking status, hypertension, and heart disease. The target variable, stroke, indicates whether a stroke occurred. These features are commonly associated with stroke risk and provide a solid foundation for identifying patterns and predicting high-risk individuals.

Cleaning the Dataset

Data cleaning included handling missing BMI values using median imputation and encoding categorical variables (such as work type and smoking status) into numerical form using one-hot encoding. The dataset was also checked for imbalance and outliers to ensure data quality. Once cleaned, the dataset was prepared for visualization and model training.

Modelling and Results

Tableau Dashboard Findings

The Tableau dashboard visualizations revealed strong relationships between key health and lifestyle factors and stroke occurrence. Age displayed a clear upward trend in stroke cases, with risk increasing significantly for individuals over 60. BMI and glucose patterns also showed noticeable differences: obese and diabetic patients experienced more stroke cases compared to those in normal health ranges.

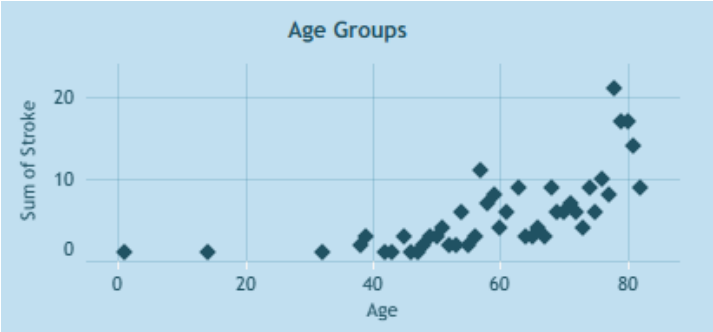


Table 1: Scatter plot of the total number of strokes that occurred at each age

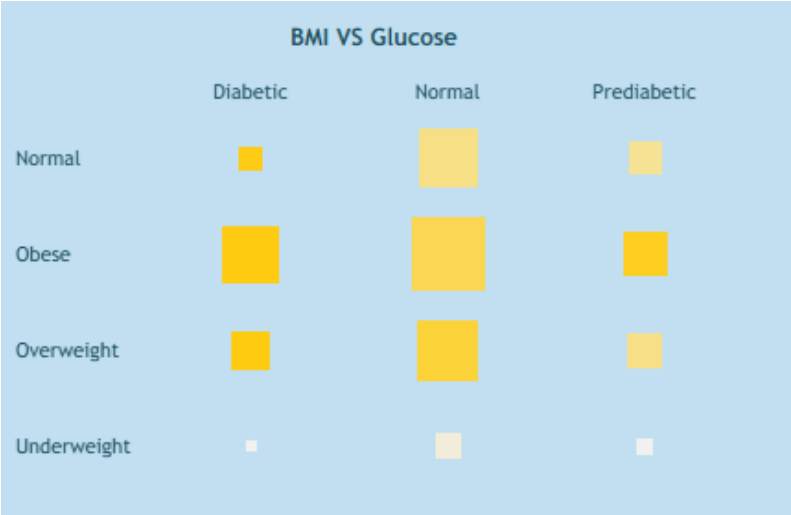


Table 2: Heat map showing the relationship between BMI category and glucose level among patients. Larger bubble sizes represent groups with a higher number of stroke-related risk factors, highlighting increased prevalence among obese and diabetic individuals.

Hypertension and heart disease strongly overlapped with stroke prevalence, confirming their role as major medical risk contributors. Lifestyle characteristics such as smoking history

demonstrated higher stroke representation among individuals who were current or former smokers.

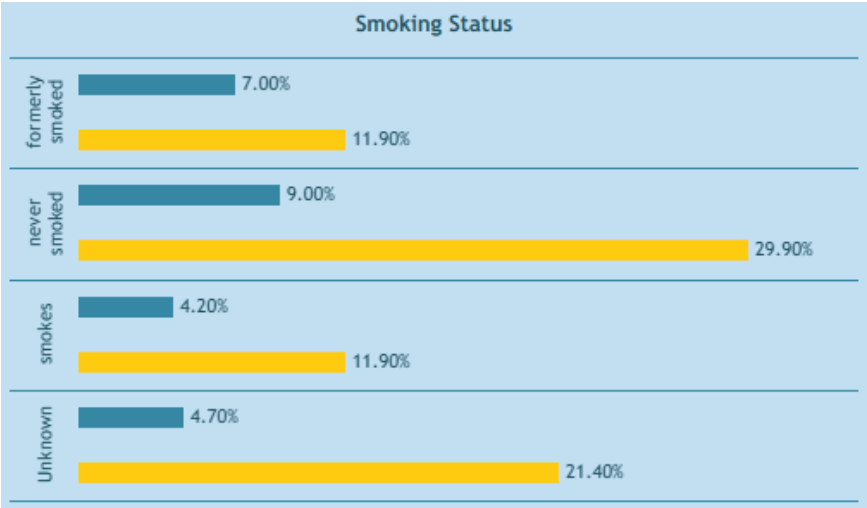


Table 3: Bar chart comparing smoking categories between stroke and non-stroke patients

Demographic visuals indicated that marital status and work type also played a role, with married individuals and private sector workers making up a large share of patients who experienced strokes. Urban residence showed slightly higher representation, but the difference was modest. Overall, the dashboard provided a clear picture of high-risk groups and supported the need for targeted health interventions for individuals with multiple overlapping conditions.

Machine Learning Findings

Logistic Regression was tested first and achieved an accuracy of 0.75, but it struggled to correctly identify stroke-positive patients due to the severe class imbalance in the dataset where only a small portion of cases represented actual strokes.

To improve detection, a Random Forest model was trained, which reached a higher accuracy of 0.95. However, this model initially still failed to classify patients who were at risk, continuing to favor the majority class. After applying SMOTE to oversample stroke cases during training, the model’s recall improved significantly while maintaining its high overall accuracy.

Model evaluation further supported this improvement. A confusion matrix confirmed that the enhanced Random Forest model successfully identified multiple true stroke cases that earlier models had missed entirely. The ROC curve also demonstrated strong discriminative ability, with an AUC score of 0.78, indicating reliable separation between high-risk and low-risk patients.

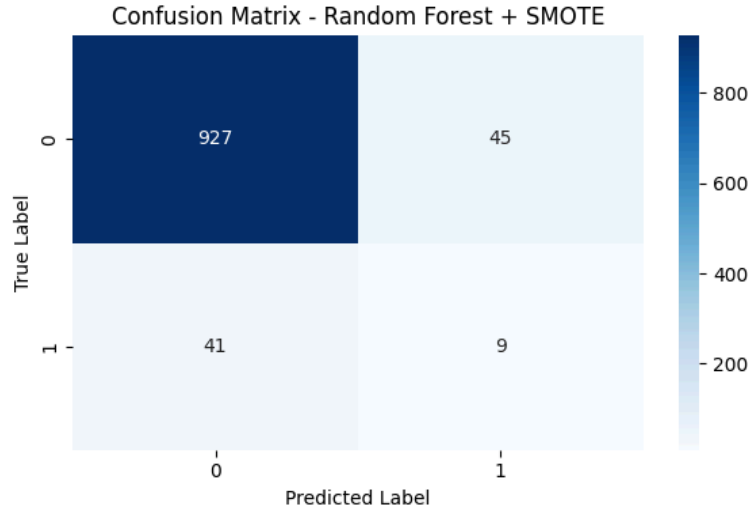


Table 4: Confusion matrix that Shows how accurately the model classified stroke and non-stroke patients.

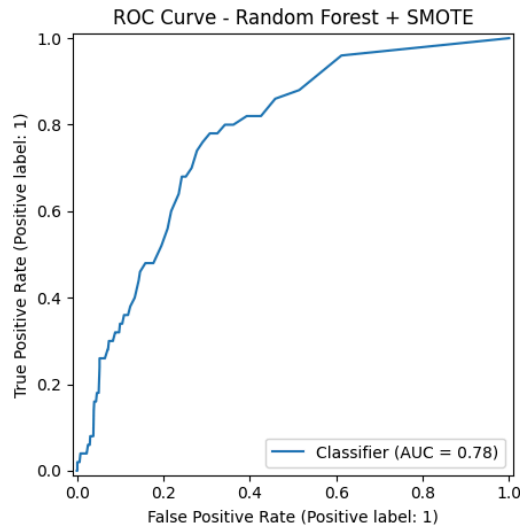


Table 5: ROC curve that shows the model's ability to distinguish between patients at high and low risk of stroke.

Feature importance analysis revealed that age was the most influential predictor of stroke risk, followed by average glucose level and BMI. Lifestyle and demographic factors, such as smoking status and residence type, contributed less to the model but still played a noticeable role. Collectively, these results show that machine learning can support early stroke risk identification when class imbalance is properly addressed and medically relevant features are incorporated.

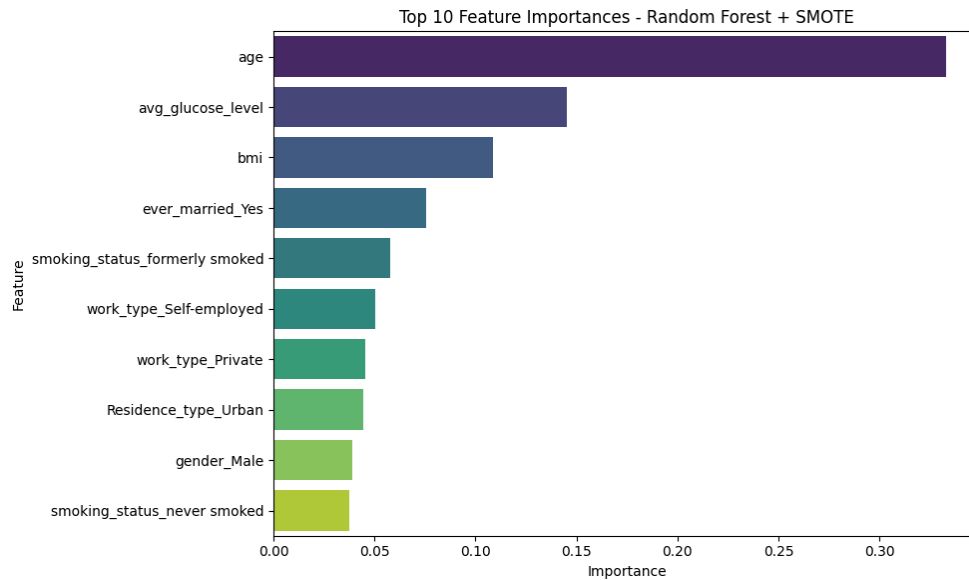


Table 6: Bar graph showing the most important features that affected the training and testing

Conclusion

Based on the analysis, the dataset revealed clear trends in patient health factors and stroke prevalence. Older adults, particularly those aged 60 and above, appeared significantly more at risk, and stroke occurrence was more common among individuals with hypertension, heart disease, or elevated glucose levels. While BMI also played a role, the combination of obesity and diabetic glucose levels showed the strongest association with stroke. Some patient information, such as detailed medical history or lifestyle habits, was limited in the dataset, which may have impacted the depth of certain findings.

One of the primary challenges in this project was the strong class imbalance, since only a small percentage of patients had experienced a stroke. This initially made it difficult for machine learning models to correctly identify high-risk individuals. To overcome this, I applied SMOTE to better represent stroke cases during training, which significantly improved predictive performance. A major success of this work was the ability to analyze health patterns using both Tableau and Python, and to successfully build a model that can support stroke risk identification while transforming these insights into clear, meaningful visualizations for users.

References

1. Fedesoriano. (2021, January 26). *Stroke prediction dataset*. Kaggle.
<https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>